ROBERTO GUAGNELLI GARCÍA

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Mechanical Engineering | Aerospace Concentration | Mechanical Design, Analysis & Manufacturing
Spanish and Mexican citizen | Authorized to work in the EU | Available after July 2027; open to relocation

EDUCATION

Tecnológico de Monterrey, Campus Estado de México Expected Jul 2027
B.S. Mechanical Engineering, Concentration in Aeronautics | **Academic average: 91/100**

PROFESSIONAL EXPERIENCE

GE Aerospace | Designer Intern, Product Definition Aug 2026 – Present
RISE Open Fan Program

- Developed 70+ support brackets and completed 20+ harness-routing revisions in Siemens NX, considering manufacturability, installation and maintainability.
- Resolved mechanical interfaces involving clearances, interferences and minimum cable bend radii; prepared product-definition drawings and datum-based tolerance tables.

GE Aerospace | Lean Summer Experience Intern Jun 2026 – Jul 2026
Integrated Performance & Operability (IP&O)

- Implemented visual planning and KPI dashboards with program leads; meeting leaders measured approximately 20% shorter meetings. Supported process automation in Python.

ENGINEERING LEADERSHIP & SELECTED PROJECTS

Borregos CEM AeroDesign | Tecnológico de Monterrey Sep 2023 – Present
Design Captain / Engineering Coordinator (now Engineering & Leadership Advisor) *Aug 2024 – Present*

- Led five designers and coordinated aircraft integration across a 40+ member team. Personally modeled 300+ CAD parts per aircraft within assemblies of 400+ components, from structures and mechanisms to systems installation.
- **Wingull (2025):** Designed removable joints for a 4.2 m wing under a 1.2 m transport limit; supported an 18 kg distributed-load test. The aircraft completed payload-carrying flight.
- **Latías (2026):** Developed longitudinal flight-dynamics models using Python and XFLR5, with stability derivatives, state-space eigenvalues, phugoid/short-period analysis and a revised loaded-CG target to recover modeled stability.

Manufacturing Coordinator | Hyperion *Sep 2023 – Aug 2024*

- Coordinated eight builders; designed and fabricated spring-loaded aluminum nose landing gear, fitted carbon-fiber skin and introduced an assembly spacer to correct rib misalignment.
- Supported six competition flights; demonstrated 3 kg payload. The team achieved **2nd overall** and **3rd in design report** at SAE Aero Design México 2024.

Computational Aerodynamics | Individual Engineering Study 2026

- Built a 3D wing CFD workflow in ANSYS Fluent (geometry, mesh, SST k–ω, steady/transient solving and post-processing); compared aerodynamic trends with XFLR5 and used findings to guide wing incidence and tip-design decisions.

Extruder Screw Failure Analysis | Academic Team Project 2026

- Investigated a fractured tool-steel extrusion screw using hardness testing, metallography, SEM/EDS and torsional stress calculations; evaluated competing failure hypotheses against physical evidence.

ADDITIONAL DIGITAL PRODUCTION EXPERIENCE

3D Asset Development & Interactive Content | Selected Collaborations 2020 – 2026

- Created **100+** 3D models and textures for Razzleberries' licensed *Hello Kitty and Friends* Minecraft content, following brand guidelines; produced **25+** detailed characters/variants for an independent Bloodborne-inspired project.
- Developed Minecraft Bedrock entities, animation behaviors and interactive mechanics using Blockbench, Blender and JSON-based configurations for additional projects.

TECHNICAL SKILLS & CERTIFICATIONS

CAD & analysis: Siemens NX, SOLIDWORKS, Fusion 360, ANSYS Fluent, XFLR5, AVL; assemblies, drawings, CFD, flight dynamics.
Programming & manufacturing: Python, MATLAB, Excel, Power BI, JSON; composites, 3D printing, laser cutting, forming, aluminum welding.
Additional: Blender, Blockbench | **CSWP certification** | Spanish (native), English (C2; IELTS 8.5), German (A2).